CSC-343: Midterm Exam

(Propositional Logic Review Questions)

- **Show all work** for full credit. Answers without supporting steps or reasoning will receive little or no credit.
- Present your work in a **clear, logical, and organized manner**. Do not scatter steps with arrows or disconnected notes.
- **Clearly indicate your final answers** (e.g., box or highlight them).
- For proofs or problems, state all assumptions explicitly and justify each step.
- Only non-graphing calculators are permitted. No other electronic devices (including cell phones) are allowed.
- Use a pencil or erasable pen for your responses.
- For questions requiring responses that are not computational, you must use complete sentences.
- Any extraneous commentary about exam content will result in a reduced professionalism grade. If you have concerns, ask clarifying questions instead.
- Scratch paper is for your use only. It will be collected but not reviewed. Transfer all final work to this exam.

By signing below, I certify that the work on this exam is my own without the use of external resources.

_____ _____

Printed Name Signature

Short Answer

1. Explain the difference between an argument being *valid* and an argument being *sound*. Why does this distinction matter for AI systems?

2. Why is enumeration (truth table checking) considered an intractable inference procedure? What is its time complexity?

3. Describe the main difference between forward chaining and backward chaining. In what scenario would you prefer backward chaining over forward chaining?

4. What does it mean for an inference system to be *complete*?

True or **False**

1. An argument can be valid even if all of its premises are false.

2. If a propositional formula φ is satisfiable, then $\sim\varphi$ must be unsatisfiable.

3. If a propositional formula φ is a tautology, then $\sim\varphi$ must be unsatisfiable.

4. Forward chaining is guaranteed to terminate on any finite knowledge base with a finite set of propositional symbols.

5. Backward chaining is more efficient than forward chaining when the goal can be proven using only a small subset of the rules in the knowledge base.

6. The resolution rule is sound because whenever both parent clauses are true, the resolvent must also be true.

7. If an inference system is complete, it must also be sound.

8. The formula $(P \Rightarrow Q)$ is logically equivalent to $(\sim Q \Rightarrow \sim P)$.

9. In the resolution refutation procedure, if we fail to derive the empty clause after exhausting all possible resolutions, we can conclude that the query is entailed by the knowledge base.

10. A formula in Conjunctive Normal Form (CNF) that contains the clause $(P \vee \sim P)$ can be simplified by removing that clause entirely without changing the formula's satisfiability.

11. Modus ponens alone (without any other inference rules) forms a complete inference system for propositional logic.

Other Questions

1. Suppose your knowledge base $KB = \{A \vee B, \sim A \vee C, \sim B \vee D, \sim C \vee D\}$. Use the resolution to prove $KB \vDash D$. Be clear about each step and explain why the result follows.

2. Convert the $(P \Leftrightarrow Q) \Rightarrow (R \wedge \sim(S \vee P))$ to Conjunctive Normal Form (CNF). Show all steps clearly, indicating which transformation rule you apply at each step.

3. Given the following knowledge base:

Facts: $\{A, B\}$

Rules:

1. $A \wedge B \rightarrow C$

2. $C \rightarrow D$
3. $B \wedge D \rightarrow E$
4. $A \wedge E \rightarrow F$
5. $C \wedge F \rightarrow G$
6. $D \wedge E \rightarrow H$

(a) Execute the forward chaining algorithm to determine all facts that can be derived. Make sure to specify the current fact database after each iteration.

(b) Using backward chaining, determine whether G can be proven from the knowledge base. Draw a directed hypergraph showing the derivation.

4. For the knowledge base below, does $KB \vDash T$? Be clear in all your steps and your resolvents.

Knowledge Base:

- $P \vee Q$
- $\neg P \vee R$
- $\neg Q \vee R$
- $\neg R \vee S$
- $\neg S \vee T$

5. An inference algorithm is *sound* if it only derives sentences that are actually entailed by the knowledge base.

(a) Define soundness formally using the symbols $\vdash$ (derives) and $\vDash$ (entails).

(b) A student builds a forward chaining system but introduces a bug: the algorithm sometimes fires a rule when only *some* of its premises are satisfied, rather than requiring *all* premises. Explain why this bug compromises soundness and give a concrete example demonstrating the problem.

6. An inference algorithm is *complete* if it can derive every sentence that is entailed by the knowledge base. A *Horn clause* is a disjunction of literals with at most one positive literal. When written in implication form, a *definite clause* (a Horn clause with exactly one positive literal) has the structure $A_1 \wedge A_2 \wedge \ldots \wedge A_n \Rightarrow B$, where all literals are positive.

(a) Define completeness formally using the symbols $\vdash$ (derives) and $\vDash$ (entails).

(b) Resolution is complete for propositional logic, but forward chaining is only complete for knowledge bases consisting of Horn clauses. Give an example of a simple knowledge base and query where forward chaining fails to derive an entailed conclusion and explain why.

7. Forward chaining and backward chaining are restricted to knowledge bases containing only Horn clauses (definite clauses), which do not permit negations. Explain why allowing negated literals in these clauses would be problematic for these algorithms.